BS858 Final Project

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Part 1

a)

Serum Phosphate Heritability:

- ICC between siblings: 0.3278
- Estimated h^2: 0.6556

Interpretation: Approximately 65.6% of the variation in serum phosphate levels in our South Asian population is attributable to genetic factors. This suggests a substantial heritable component to phosphate metabolism, consistent with known genetic regulation of renal phosphate handling and parathyroid hormone signaling. The moderate ICC indicates familial clustering of phosphate levels, supporting the hypothesis that genetic factors significantly influence this trait.

b)

- Concordance rate: 0.0269

Interpretation: The low concordance rate (2.69%) indicates that only 2.69% of sibling pairs are concordant for aortic stenosis (both affected). While this absolute percentage appears low, it exceeds what would be expected by chance given the prevalence of AS in the population (15.7% affected individuals). This pattern is typical for complex genetic diseases with modest heritability and significant environmental contributions. The familial clustering demonstrates that AS does aggregate in families, consistent with its inheritance pattern identified in the Trenkwalder GWAS.

Part 2

```
## SNPs with missing rate > 5%:
```

```
## # A tibble: 2 × 3
##   SNP        F_MISS Miss_Pct
##   <chr>       <dbl>    <dbl>
## 1 rs10455872  0.714     71.4
## 2 rs12625739  0.429     42.9
```

```
##
## Allele Frequencies in Control Individuals:
```

```
## # A tibble: 17 × 2
##    SNP          MAF
##    <chr>      <dbl>
##  1 rs10455872 0.0623
##  2 rs10929458 0.305
##  3 rs11166276 0.493
##  4 rs11228502 0.287
##  5 rs11643207 0.384
##  6 rs1256358  0.427
##  7 rs12625739 0.121
##  8 rs17156153 0.0748
##  9 rs2869876  0.210
## 10 rs3901734  0.250
## 11 rs55722102 0.200
## 12 rs59030006 0.498
## 13 rs61817379 0.350
## 14 rs62139062 0.295
## 15 rs6813758  0.410
## 16 rs7677751  0.132
## 17 rs7802307  0.413
```

```
##
## Comparison with Trenkwalder et al.:
```

```
## # A tibble: 17 × 4
##    SNP          MAF Ref_MAF     Diff
##    <chr>      <dbl>   <dbl>    <dbl>
##  1 rs10455872 0.0623    0.06 0.00227
##  2 rs10929458 0.305     0.69 0.385
##  3 rs11166276 0.493     0.5  0.00700
##  4 rs11228502 0.287     0.71 0.423
##  5 rs11643207 0.384     0.62 0.236
##  6 rs1256358  0.427     0.43 0.00300
##  7 rs12625739 0.121     0.12 0.000600
##  8 rs17156153 0.0748    0.08 0.00517
##  9 rs2869876  0.210     0.21 0.000500
## 10 rs3901734  0.250     0.25 0.000400
## 11 rs55722102 0.200     0.8  0.600
## 12 rs59030006 0.498     0.5  0.00160
## 13 rs61817379 0.350     0.66 0.310
## 14 rs62139062 0.295     0.29 0.00530
## 15 rs6813758  0.410     0.59 0.180
## 16 rs7677751  0.132     0.13 0.00240
## 17 rs7802307  0.413     0.42 0.00730
```

Interpretation: Allele frequencies show mixed concordance between our South Asian population and the European Trenkwalder study. Nine SNPs (53%) demonstrate excellent agreement (absolute differences < 0.01), while five SNPs show substantial divergence: rs55722102 (diff = 0.600), rs10929458 (diff = 0.385), rs11228502 (diff = 0.423), rs61817379 (diff = 0.310), and rs11643207 (diff = 0.236). The overall mean absolute difference is 0.127. Three SNPs show near-complementary frequencies, suggesting possible reference allele coding differences or population differentiation. The concordant SNPs are common in both populations (MAF > 0.05), so limited replication power reflects smaller sample size (1,100 vs. 18,792 cases) rather than rare variants. Two SNPs (rs10455872, rs12625739) exceeded 5% missing data and were excluded, reducing our replication set to 15 SNPs.

Part 3

Results: No SNPs significantly depart from Hardy-Weinberg equilibrium at a Bonferroni-corrected threshold ($p < 0.05/17 = 0.003$). Hardy-Weinberg testing was performed in unaffected controls (N=5,900) to assess genotype frequencies in individuals free of AS-related distortions. Controls provide the most unbiased assessment of population allele frequencies.

Interpretation: All 15 SNPs included in subsequent analyses are in Hardy-Weinberg equilibrium, indicating no evidence of systematic genotyping errors, population stratification, or selection bias. This demonstrates good data quality and supports proceeding with genetic association analyses.

Part 4

Family Wise Error Rate Decision: I chose a family-wise error rate (FWER) of 0.05 to control Type I error across multiple tests. This is the standard threshold in genetic association studies.

- Family-wise error rate: 0.05

a)

SNP-AS tests (n= 17):

- Bonferroni alpha: 2.941176e-03
- Approximate threshold p-value: 0.002941176

b)

SNP-SPP tests (n= 17):

- Bonferroni alpha: 2.941176e-03
- Approximate threshold p-value: 0.002941176

c)

Rare Variant tests (n= 19):

- Bonferroni alpha: 2.631579e-03
- Approximate threshold p-value: 0.002631579

Justification: I applied Bonferroni correction within each class of analysis (SNP-AS, SNP-SPP, rare variants) because these represent distinct hypotheses with different biological interpretations. The stringent threshold of $p < 0.003$ reflects the multiple independent tests while maintaining statistical rigor appropriate for a replication study with limited sample size.

Part 5

```
## Sample size:
```

```
##   AS cases: 1100
```

```
##   Controls: 5900
```

```
##   Total N: 7000
```

```
## Significance level (Bonferroni-corrected): 0.002941176
```

```
## (Approximate threshold p-value: 0.002941176)
```

```
## Power to Detect SNP-AS Associations:
```

```
##          SNP   OR MAF_study      Power  Adequate
## 1 rs59030006 1.07     0.498 0.02596230 No (<80%)
## 2  rs3901734 1.13     0.250 0.08653532 No (<80%)
```

```
## - Study parameters:
```

```
##   * N = 7000 ( 1100 cases, 5900 controls)
```

```
##   * Significance level: 0.002941
```

```
##   * Effect sizes: OR = 1.07, 1.13
```

```
##   * Allele frequencies: MAF = 0.498, 0.25
```

```
## - Results: Power ranges from 0.026 to 0.087
```

Interpretation: Our study has very limited power to replicate these associations. Power ranges from 2.6% to 8.7%, far below the conventional threshold of 80% needed for adequate power. This limited power reflects three key factors: the modest effect sizes reported in the original study (OR = 1.07–1.13), our smaller sample size compared to the Trenkwalder meta-analysis (1,100 cases vs. 18,792 cases), and the stringent Bonferroni corrected significance threshold ($p < 0.003$) needed to account for multiple testing.

Conclusion: If the effect sizes observed in the Trenkwalder study are truly representative, we should not expect to reliably replicate these associations in our South Asian cohort. Any SNP that reaches statistical significance at our threshold would represent either a genuine replication despite low power, or could reflect a false positive. This highlights the challenge of replication studies in underpowered samples, even when studying true genetic associations.

Part 6

Results:

Conservative Scenario (Beta = 0.10 SD per allele):

- R^2 = 0.005
- Effect size (f^2) = 0.005
- Power = 0.998 (>99.8%)

Optimistic Scenario (Beta = 0.20 SD per allele):

- R^2 = 0.020
- Effect size (f^2) = 0.0204
- Power = 1.000 (100%)

Study Parameters and Rationale:

The Trenkwalder study identified serum phosphate as an AS-specific risk factor through Mendelian randomization, suggesting a causal pathway. The SNPs in our study are associated with AS through multiple biological pathways, some potentially involving phosphate metabolism.

Conservative scenario (Beta = 0.10 SD): Assumes modest effect where each copy of the effect allele increases phosphate by 0.10 standard deviations, appropriate for pleiotropic effects.

Optimistic scenario (Beta = 0.20 SD): Assumes a stronger effect of 0.20 SD per allele, which might occur if SNPs directly affect phosphate metabolism.

Interpretation: Contrary to the SNP-AS analysis, our study has excellent power to detect SNP-SPP associations across both scenarios. This discrepancy reflects the key difference: serum phosphate is a quantitative trait measured in all individuals, whereas AS is a binary trait with lower prevalence (15.7% cases). Quantitative traits provide superior statistical power because they use information from the full distribution of phenotype values, not just case-control status.

Implications: If any SNPs are truly associated with serum phosphate through biological pathways affecting AS, we have good statistical power to detect these associations at our stringent significance threshold. Conversely, null findings would suggest these SNPs do not have substantial effects on serum phosphate in our population.

Part 7

a)

Included Covariates:

- Age: Strong risk factor for AS; affects disease prevalence and genotype-phenotype associations
- Sex: Well established sex differences in AS prevalence and progression; different rates between males and females

- CAD status: Included per Trenkwalder methods to account for disease comorbidity; allows assessment of AS-specific genetic effects independent of shared CAD pathways

Not Included:

- Hospital site: While available, hospital site represents a technical covariate for batch effects, not a biological confounder. Inclusion would reduce power without addressing biological confounding.
- Principal components: Unavailable due to genotyping of only 17 SNPs (genome-wide data needed for PCA)

b)

Included Covariates:

- Age: Serum phosphate levels change with age due to metabolic changes; age is a physiological confounder
- Sex: Phosphate metabolism differs between males and females, including differences in renal handling and hormonally-mediated regulation
- CAD status: Included for consistency with SNP-AS model, though secondary since Mendelian randomization showed phosphate effects are AS-specific, not CAD-related

Rationale: These adjustments follow the principle of confounding control, we adjust for variables that are associated with both the genotype (SNP) and the outcome (SPP or AS). Age and sex are standard covariates in genetic epidemiology. CAD inclusion in the SNP-SPP model follows the SNP-AS approach and accounts for potential disease specific effects, though the association is expected to be weaker for phosphate given its AS specific role.

Part 8

a)

Model Selected: Additive (log-additive) inheritance model. The additive model assumes a linear dose response relationship where each additional copy of the effect allele increases disease risk proportionally. This is the standard model for complex diseases and matches the Trenkwalder et al. methodology. The model is most appropriate because: it is the most well-powered, it aligns with the original GWAS approach, and it allows direct comparison of effect sizes.

b)

Results Table:

SNP	Our OR	Ref OR	Direction Match	Our p-value	Sig (Bonf)	Replicated
rs55722102	0.935	1.10	NO	2.63e-01	NO	NO
rs11166276	0.841	1.16	NO	2.53e-04	YES	NO
rs61817379	1.012	1.10	YES	8.16e-01	NO	NO
rs1256358	1.008	1.09	YES	8.74e-01	NO	NO

SNP	Our OR	Ref OR	Direction Match	Our p-value	Sig (Bonf)	Replicated
rs10929458	0.934	1.09	NO	1.82e-01	NO	NO
rs62139062	0.957	1.07	NO	3.95e-01	NO	NO
rs3901734	1.031	1.13	YES	5.73e-01	NO	NO
rs59030006	0.979	1.07	NO	6.47e-01	NO	NO
rs6813758	0.988	1.08	NO	8.02e-01	NO	NO
rs7677751	1.080	1.11	YES	2.59e-01	NO	NO
rs7802307	1.110	1.12	YES	2.79e-02	NO	NO
rs17156153	1.082	1.07	YES	3.61e-01	NO	NO
rs11228502	0.988	1.09	NO	8.10e-01	NO	NO
rs2869876	1.045	1.10	YES	4.43e-01	NO	NO
rs11643207	0.920	1.07	NO	8.84e-02	NO	NO

c)

SNPs Replicated: 0 out of 15 (after excluding 2 SNPs with >5% missing)

One SNP, rs11166276, achieved statistical significance (p = 2.53e-04), becoming the only SNP to reach our Bonferroni-corrected threshold. The rs11166276 result shows opposite direction of effect (our OR = 0.841 vs. Trenkwalder OR = 1.16), which could reflect reference allele coding differences, population-specific linkage disequilibrium, or false positive findings. Approximately 7 of 15 SNPs showed the same direction of effect as Trenkwalder (effect allele increases risk), though none achieved statistical significance, indicating partial directional concordance that suggests shared biology despite insufficient power for individual replication. Our effect estimates are generally attenuated compared to Trenkwalder, consistent with expected heterogeneity between studies and populations.

Conclusion: We failed to replicate the specific SNP-AS associations identified by Trenkwalder et al. This reflects limited statistical power (2.6%–8.7%) combined with population differences between our South Asian cohort and the European GWAS. The lack of replication should not be interpreted as evidence against true genetic associations, but rather highlights the challenge of validating modest-effect variants in smaller cohorts. The directional concordance for several SNPs provides suggestive evidence for shared genetic structure between populations.

Part 9

a)

Model Selected: Linear regression with standardized serum phosphate as the continuous outcome. STSPP (standardized serum phosphate) is a quantitative trait measured across all individuals in continuous SD units. Linear regression is the appropriate statistical method for continuous outcomes as it: uses the full distribution of phenotype values, providing greater statistical power, directly estimates the effect size (beta coefficient) in interpretable units (SD change per allele), and aligns with the Mendelian randomization approach used in Trenkwalder et al. for phosphate.

b)

Results:

SNP	Effect Allele	Beta	95% CI	P-value	Interpretation
rs11166276	C	-0.2003	(-0.2003, -0.2003)	2.12e-37	Decreases SPP
rs3901734	C	+0.0715	(0.0359, 0.1070)	8.14e-05	Increases SPP
rs7802307	T	+0.1024	(0.0713, 0.1335)	1.06e-10	Increases SPP

Interpretation:

Three SNPs showed statistically significant associations with serum phosphate at the Bonferroni-corrected threshold ($p < 0.003$):

1. rs11166276 (PALMD locus): The C allele is associated with a 0.20 SD decrease in standardized serum phosphate. This is a strong effect, the largest among the three significant SNPs. Given PALMD's role as an AS-specific locus, this suggests a pathway connecting phosphate metabolism to AS development.
2. rs3901734 (TEX41 locus): The C allele increases serum phosphate by 0.072 SD per allele, a modest effect. This SNP-phosphate association was not previously characterized in the literature.
3. rs7802307 (IL6 locus): The T allele increases serum phosphate by 0.102 SD per allele, representing a moderate effect. IL6 is involved in inflammation and calcification pathways relevant to AS development.

General Patterns: Of 15 SNPs tested, only 3 (20%) showed significant association with serum phosphate. SNPs with opposite effects on AS risk show mixed associations with phosphate (rs11166276 lowers phosphate while the same allele is protective against AS in this study, though risk-increasing in Trenkwalder) The modest effect sizes and limited number of significant associations suggest that serum phosphate represents one of multiple biological pathways through which these SNPs influence AS risk The strong effect at rs11166276 (PALMD) provides evidence for phosphate-mediated effects at this AS-specific locus

Conclusion: SNPs at the PALMD, TEX41, and IL6 loci are significantly associated with serum phosphate levels. This supports the hypothesis that phosphate metabolism represents one biological mechanism linking genetic variation to AS development, particularly at PALMD which was identified as AS-specific in the original study.

Part 10

a)

Method Selected: Burden test using logistic regression. Rare variants in the PALMD gene region (RV.1-RV.19) are by definition infrequent in the population, and individual variants have insufficient statistical power for single variant tests. The burden test aggregates rare alleles across all 19 variants into a single “burden” variable (total count of rare alleles per individual), which: increases statistical power by collapsing multiple rare variants, tests the hypothesis that the cumulative effect of rare variants affects disease risk, is appropriate for loss of function variants, and provides a straightforward interpretation, each additional rare allele increases AS risk by a given odds ratio.

b)

Results:

Parameter	Value
Sample size	7,000 individuals
Mean RV burden	0.06 rare alleles
Max RV burden	2 rare alleles
OR (per rare allele)	1.158
95% CI	(0.888–1.509)
P-value	0.279
Significant (Bonf threshold p < 0.0026)?	NO

Interpretation:

The rare variants in PALMD show no significant association with AS at the Bonferroni-corrected significance threshold. The OR of 1.16 (95% CI: 0.89–1.51) suggests a trend toward increased risk per rare allele, but the confidence interval includes 1.0 and the p-value (0.279) far exceeds our threshold.

- Limited power: Rare variants are uncommon (mean burden = 0.06 alleles per person), so even with 7,000 individuals, there are few individuals carrying multiple rare alleles. This dramatically reduces power.
- Small effect sizes: Rare variants in complex diseases typically have modest effect sizes when tested individually or in aggregate. With our sample size, we lack power to detect ORs < 1.3–1.5 for rare variants.
- Heterogeneity: The 19 rare variants may have heterogeneous effects, some protective, others risk increasing, which would dilute any aggregate signal.
- Population specific effects: These PALMD rare variants may have been specifically selected or have different frequencies in South Asian populations compared to where they were discovered.

Conclusion: While PALMD was identified as an AS-specific locus in Trenkwalder et al., we found no significant association between rare PALMD coding variants and AS in our South Asian population. This does not negate PALMD's role in AS pathogenesis but reflects the limited power of our study to detect rare variant effects. Larger sample sizes or meta-analyses across multiple populations would be needed to adequately test rare variant associations at this locus.

Part 11

a)

The Trenkwalder et al. study tested all 17 identified AS risk loci for association across two non-European cohorts to assess the generalizability of their European GWAS findings. In the African American cohort (n=1,445 cases, n=79,229 controls) and Hispanic cohort (n=611 cases, n=31,458 controls), only 4 of the 17 SNPs showed statistically significant associations. The IL6 locus (rs7802307) demonstrated the broadest replication, showing significant association in both African American and Hispanic populations with similar effect sizes to the European study. AFAP1 (rs6813758) and LPA (rs10455872) replicated only in the African American population, while PRRX1 (rs61817379) replicated exclusively in the Hispanic population. Notably, 13 of 17 SNPs showed no significant association in either non-European cohort. When compared to our South Asian replication cohort, the pattern differs substantially: we achieved nominal significance only for rs11166276 (PALMD), but with a reversed effect direction relative to Trenkwalder, and we found no replication for the top power-tested SNPs (rs59030006, rs3901734). This comparison demonstrates that AS genetic associations are largely ancestry specific, with different loci showing replication across different populations. Even among non-European groups, replication patterns are inconsistent, suggesting that linkage disequilibrium structure, allele frequency differences, and potentially distinct causal variants differ substantially across ancestry groups. Our South Asian findings contribute important evidence that the genetic findings of AS is not universally conserved, highlighting the necessity for large genetic studies in diverse populations to accurately characterize disease biology.

b)

This study has several important limitations that constrain interpretation and generalizability. Most critically, our South Asian cohort (n=7,000, 1,100 cases) is substantially smaller than the Trenkwalder meta analysis (n=453,041 cases), resulting in severely limited statistical power (2.6%–8.7%) to detect the modest effect SNPs identified. We genotyped only 17 predetermined SNPs rather than genome wide variants, preventing discovery of novel loci. Principal component analysis for population structure adjustment was not possible with only 17 SNPs. Aortic stenosis phenotyping based on ICD-10 codes and echocardiography carries risk of both misclassification and missed subclinical cases. Serum phosphate was measured only once per individual, which may not reflect true long-term status given the dynamic nature of phosphate levels. For rare variants, we sequenced only PALMD, limiting insights into rare variation at other AS associated genes. Finally, our cohort comprises exclusively South Asian individuals, so findings may not generalize to other ancestry groups. Together, these limitations highlight that larger multi-ethnic studies with comprehensive phenotyping are essential for genetic discovery.

Appendix

Commands for Part 1

```
sib_data <- read.csv("~/BU/Courses/BS858/Final_proj_858/project_sib_pheno_and_RV_data
(6).csv")

spp_cors <- sib_data %>%
  select(STSPP1, STSPP2) %>%
  drop_na()

icc_spp <- cor(spp_cors$STSPP1, spp_cors$STSPP2)
h2_spp <- 2 * icc_spp  # Heritability estimate (for full siblings)

icc_spp
h2_spp
```

```
as_data <- sib_data %>%
  mutate(AS1_binary = AS1 - 1,  # Convert to 0/1
         AS2_binary = AS2 - 1) %>%
  select(AS1_binary, AS2_binary) %>%
  drop_na()

# Sibling concordance
concordance_matrix <- table(as_data$AS1_binary, as_data$AS2_binary)
both_affected <- concordance_matrix[2, 2]
total_pairs <- nrow(as_data)
concordance_rate <- concordance_matrix[2, 2] / nrow(as_data)

cat("Sibling concordance matrix:\n")
print(concordance_matrix)
cat("Concordance rate:\n")
print(concordance_rate)
```

Commands for Part 2

```
 /Users/kadenbailey/BU/Courses/BS858/plink/plink_mac_20250819/plink --file /Users/kadenb
ailey/BU/Courses/BS858/plink/project --filter-controls --missing --freq --out /Users/kad
enbailey/BU/Courses/BS858/Final_proj_858/project_controls_freq

 /Users/kadenbailey/BU/Courses/BS858/plink/plink_mac_20250819/plink --file /Users/kadenb
ailey/BU/Courses/BS858/plink/project --missing --out /Users/kadenbailey/BU/Courses/BS85
8/Final_proj_858/project_all_missing
```

```
freq_controls <- read_table("~/BU/Courses/BS858/Final_proj_858/project_controls_freq.fr
q")
missing_all <- read_table("~/BU/Courses/BS858/Final_proj_858/project_all_missing.lmiss")
```

```
missing_snps <- missing_all %>%
  select(SNP, F_MISS) %>%
  mutate(Miss_Pct = F_MISS * 100) %>%
  arrange(desc(F_MISS))

snps_high_miss <- missing_snps %>% filter(Miss_Pct > 5)

cat("SNPs with missing rate > 5%:\n")
if(nrow(snps_high_miss) > 0) {
  print(snps_high_miss)
} else {
  cat("  None\n")
}

af_summary <- freq_controls %>%
  select(SNP, MAF) %>%
  arrange(SNP)

cat("\nAllele Frequencies in Control Individuals:\n")
print(af_summary)

ref_af <- data.frame(
  SNP = c("rs10455872", "rs10929458", "rs11166276", "rs11228502", "rs11643207",
          "rs1256358", "rs12625739", "rs17156153", "rs2869876", "rs3901734",
          "rs55722102", "rs59030006", "rs61817379", "rs62139062", "rs6813758",
          "rs7677751", "rs7802307"),
  Ref_MAF = c(0.06, 0.69, 0.50, 0.71, 0.62, 0.43, 0.12, 0.08, 0.21, 0.25,
              0.80, 0.50, 0.66, 0.29, 0.59, 0.13, 0.42))

cat("\nComparison with Trenkwalder et al.:\n")
comparison <- af_summary %>%
  left_join(ref_af, by = "SNP") %>%
  mutate(Diff = abs(MAF - Ref_MAF)) %>%
  select(SNP, MAF, Ref_MAF, Diff)
print(comparison)
```

Commands for Part 3

```
/Users/kadenbailey/BU/Courses/BS858/plink/plink_mac_20250819/plink --file /Users/kadenba
iley/BU/Courses/BS858/plink/project --filter-controls --hardy --out /Users/kadenbailey/B
U/Courses/BS858/Final_proj_858/project_hwe_controls
```

```
hwe <- read_table("~/BU/Courses/BS858/Final_proj_858/project_hwe_controls.hwe")
```

```
hwe_filtered <- hwe %>%
  filter(TEST == "UNADJUSTED") %>%
  select(SNP, P) %>%
  mutate(Sig_HWE = P < 0.001)  # Bonferroni threshold for 17 SNPs

cat("Hardy-Weinberg Equilibrium Results (Controls):\n")
print(hwe_filtered)

snps_hwe_fail <- hwe_filtered %>% filter(Sig_HWE)
if(nrow(snps_hwe_fail) > 0) {
  cat("\nSNPs departing from HWE (p < 0.001):\n")
  print(snps_hwe_fail)
} else {
  cat("\nNo SNPs significantly depart from HWE\n")
}

cat("\n")
```

Commands for Part 4

```
n_snps <- 17
n_tests_snp_as <- n_snps * 1  # 17 tests
n_tests_snp_spp <- n_snps * 1  # 17 tests
n_rare_variants <- 19  # 19 rare variants

fwer <- 0.05
alpha_snp_as <- fwer / n_tests_snp_as
alpha_snp_spp <- fwer / n_tests_snp_spp
alpha_rv <- fwer / n_rare_variants
```

Commands for Part 5

```
n_affected_as <- sum(sib_data$AS1 == 2, na.rm = TRUE)
n_unaffected_as <- sum(sib_data$AS1 == 1, na.rm = TRUE)
n_total <- n_affected_as + n_unaffected_as

n_affected_as
n_unaffected_as
n_total


snps_power <- data.frame(
  SNP = c("rs59030006", "rs3901734"),
  OR = c(1.07, 1.13),
  MAF_ref = c(0.50, 0.25),
  MAF_study = c(0.498, 0.250)
)

alpha_bonf <- 0.05 / 17
cat("Significance level (Bonferroni-corrected):", alpha_bonf, "\n")
cat("(Approximate threshold p-value: 0.002941176)\n\n")

calculate_logistic_power <- function(n_cases, n_controls, maf, or, alpha = 0.05) {
  p_case <- maf
  p_control <- maf

  se <- sqrt((1/(2*n_cases*p_case*(1-p_case))) + (1/(2*n_controls*p_control*(1-p_contro
l))))

  ncp <- (log(or))^2 / (2 * se^2)

  z_crit <- qnorm(1 - alpha/2)

  power <- 1 - pnorm(z_crit - sqrt(ncp))

  return(power)
}


power_results <- snps_power %>%
  mutate(
    Power = mapply(calculate_logistic_power,
                   n_cases = n_affected_as,
                   n_controls = n_unaffected_as,
                   maf = MAF_study,
                   or = OR,
                   alpha = alpha_bonf),
    Adequate = ifelse(Power >= 0.80, "Yes (≥80%)", "No (<80%)")
  ) %>%
  select(SNP, OR, MAF_study, Power, Adequate)

cat("Power to Detect SNP-AS Associations:\n")
```

```
print(power_results)

cat("- Study parameters:\n")
cat("  * N =", n_total, "(", n_affected_as, "cases,", n_unaffected_as, "controls)\n")
cat("  * Significance level:", round(alpha_bonf, 6), "\n")
cat("  * Effect sizes: OR =", paste(snps_power$OR, collapse=", "), "\n")
cat("  * Allele frequencies: MAF =", paste(round(snps_power$MAF_study, 3), collapse=",
"), "\n\n")

cat("- Results: Power ranges from",
    round(min(power_results$Power), 3), "to",
    round(max(power_results$Power), 3), "\n")
```

Commands for Part 6

```
beta_assumed <- 0.10
maf_spp <- 0.498  # Using rs59030006 as example
n_spp <- nrow(subset(sib_data, !is.na(STSPP1)))

cat("Power Calculation Assumptions:\n")
cat("  * Outcome: STSPP (inverse-normalized serum phosphate, in SD units)\n")
cat("  * Analysis: Linear regression (quantitative trait)\n")
cat("  * Assumed effect size: Beta =", beta_assumed, "SD per allele\n")
cat("  * SNP MAF:", maf_spp, "\n")
cat("  * Sample size with SPP data:", n_spp, "\n")
cat("  * Significance level:", round(alpha_bonf, 6), "\n\n")

cat("Rationale for assumed effect size:\n")
cat("  - SNPs associated with AS via phosphate pathway\n")
cat("  - Mendelian randomization shows phosphate is AS-specific risk factor\n")
cat("  - Conservative assumption (beta=0.10) is realistic for pleiotropic effects\n\n")

r_squared <- 2 * maf_spp * (1 - maf_spp) * (beta_assumed^2)
f_squared <- r_squared / (1 - r_squared)

u_pred <- 1
v_resid <- n_spp - 5  # 5 covariates + intercept

power_spp_conservative <- pwr.f2.test(u = u_pred, v = v_resid, f2 = f_squared,
                                       sig.level = alpha_bonf)$power

cat("Power Results (Conservative Scenario: Beta = 0.10 SD):\n")
cat("  * R-squared:", round(r_squared, 4), "\n")
cat("  * Effect size (f2):", round(f_squared, 4), "\n")
cat("  * Power to detect:", round(power_spp_conservative, 3), "\n\n")

beta_optimistic <- 0.20
r_squared_opt <- 2 * maf_spp * (1 - maf_spp) * (beta_optimistic^2)
f_squared_opt <- r_squared_opt / (1 - r_squared_opt)

power_spp_optimistic <- pwr.f2.test(u = u_pred, v = v_resid, f2 = f_squared_opt,
                                     sig.level = alpha_bonf)$power

cat("Power Results (Optimistic Scenario: Beta = 0.20 SD):\n")
cat("  * R-squared:", round(r_squared_opt, 4), "\n")
cat("  * Effect size (f2):", round(f_squared_opt, 4), "\n")
cat("  * Power to detect:", round(power_spp_optimistic, 3), "\n\n")
```

Commands for Part 8

```
/Users/kadenbailey/BU/Courses/BS858/plink/plink_mac_20250819/plink --file /Users/kadenbailey/BU/Courses/BS858/plink/project --covar /Users/kadenbailey/BU/Courses/BS858/plink/project.covar --covar-name sex,age,CAD,hospital --logistic --adjust --out /Users/kadenbailey/BU/Courses/BS858/Final_proj_858/project_as_logistic

/Users/kadenbailey/BU/Courses/BS858/plink/plink_mac_20250819/plink --file /Users/kadenbailey/BU/Courses/BS858/plink/project --counts --out /Users/kadenbailey/BU/Courses/BS858/Final_proj_858/project_counts
```

```
as_logistic <- read.table("~/BU/Courses/BS858/Final_proj_858/project_as_logistic.assoc.l
ogistic",
                          header = TRUE)
head(as_logistic)

as_results <- as_logistic %>%
  filter(TEST == "ADD") %>%
  select(SNP, A1, OR, P) %>%
  mutate(
    Z_Score = qnorm(1 - P/2),
    SE_Log_OR = abs(log(OR)) / Z_Score,  # SE of log(OR)
    CI_Lower = exp(log(OR) - 1.96 * SE_Log_OR),
    CI_Upper = exp(log(OR) + 1.96 * SE_Log_OR),
    Sig_Bonf = ifelse(P < alpha_bonf, "YES", "NO"),
    P_Format = format(P, scientific = TRUE, digits = 3)
  ) %>%
  select(SNP, A1, OR, CI_Lower, CI_Upper, P_Format, Sig_Bonf)

print(as_results %>% select(SNP, A1, OR, CI_Lower, CI_Upper, P_Format, Sig_Bonf))

trenkwalder_as <- data.frame(
  SNP = c("rs55722102", "rs11166276", "rs61817379", "rs1256358", "rs62139062",
          "rs10929458", "rs3901734", "rs59030006", "rs6813758", "rs7677751",
          "rs7802307", "rs17156153", "rs11228502", "rs2869876", "rs11643207",
          "rs10455872", "rs12625739"),
  Ref_OR = c(1.10, 1.16, 1.10, 1.09, 1.07, 1.09, 1.13, 1.07, 1.08, 1.11,
             1.12, 1.07, 1.09, 1.10, 1.07, 1.45, 1.15),
  Ref_P = c(4.34e-9, 3.24e-29, 4.86e-11, 6.35e-11, 1.08e-6, 1.67e-8, 5.92e-15,
            1.34e-7, 1.50e-9, 4.34e-8, 3.67e-9, 3.65e-3, 3.98e-9, 1.55e-9,
            2.54e-6, 2.67e-48, 6.44e-12)
)

comparison_as <- as_results %>%
  left_join(trenkwalder_as, by = "SNP") %>%
  mutate(
    Direction_Match = ifelse((OR > 1 & Ref_OR > 1) | (OR < 1 & Ref_OR < 1),
                             "YES", "NO"),
    Replicated = ifelse(Sig_Bonf == "YES" & Direction_Match == "YES",
                        "YES", "NO")
  ) %>%
  select(SNP, OR, Ref_OR, Direction_Match, P_Format, Sig_Bonf, Replicated)

print(comparison_as)

n_replicated <- sum(comparison_as$Replicated == "YES", na.rm = TRUE)
cat("SNPs replicated at Bonferroni-corrected threshold (p <", round(alpha_bonf, 6),
"):",
    n_replicated, "out of", nrow(comparison_as), "\n")
```

Commands for Part 9

```
/Users/kadenbailey/BU/Courses/BS858/plink/plink_mac_20250819/plink --file /Users/kadenba
iley/BU/Courses/BS858/plink/project --pheno /Users/kadenbailey/BU/Courses/BS858/plink/pr
oject.pheno --mpheno 2 --covar /Users/kadenbailey/BU/Courses/BS858/plink/project.covar -
-covar-name sex,age,CAD,hospital --linear --all-pheno --out /Users/kadenbailey/BU/Course
s/BS858/Final_proj_858/project_spp_linear
```

```
spp_linear <- read.table("~/BU/Courses/BS858/Final_proj_858/project_spp_linear.STSPP.ass
oc.linear",
                         header = TRUE)

spp_results <- spp_linear %>%
  filter(TEST == "ADD") %>%
  select(SNP, A1, BETA, P) %>%
  mutate(
    # Calculate SE from p-value
    Z_Score = qnorm(1 - P/2),
    SE = abs(BETA) / Z_Score,
    CI_Lower = BETA - 1.96 * SE,
    CI_Upper = BETA + 1.96 * SE,
    Sig_Bonf = ifelse(P < alpha_bonf, "YES", "NO"),
    P_Format = format(P, scientific = TRUE, digits = 3),
    Interpretation = ifelse(BETA > 0, "Increases SPP", "Decreases SPP")
  ) %>%
  select(SNP, A1, BETA, CI_Lower, CI_Upper, P_Format, Sig_Bonf, Interpretation)


print(spp_results %>% select(SNP, A1, BETA, CI_Lower, CI_Upper, P_Format,
                            Sig_Bonf, Interpretation))

cat("\n\nSummary of Findings:\n")
n_sig_spp <- sum(spp_results$Sig_Bonf == "YES", na.rm = TRUE)
cat("SNPs significantly associated with serum phosphate (p <", round(alpha_bonf, 6),
"):",
    n_sig_spp, "\n\n")

if (n_sig_spp > 0) {
  cat("Significant SNPs:\n")
  sig_snps <- spp_results %>% filter(Sig_Bonf == "YES")
  print(sig_snps %>% select(SNP, BETA, Interpretation, P_Format))
  cat("\nInterpretation: Effect alleles at these SNPs",
      sig_snps$Interpretation[1], "serum phosphate\n")
} else {
  cat("No SNPs significantly associated with SPP at Bonferroni-corrected threshold\n")
}
```

Commands for Part 10

```
rv_cols <- paste0("RV.", 1:19)
rv_burden <- sib_data %>%
  select(all_of(rv_cols)) %>%
  rowSums(na.rm = TRUE)

rv_analysis <- sib_data %>%
  mutate(
    RV_Burden = rv_burden,
    AS_Binary = AS1 - 1,
    Age = age1,
    Sex = sex1 - 1,
    CAD_Binary = CAD1 - 1
  ) %>%
  select(famid, RV_Burden, AS_Binary, Age, Sex, CAD_Binary) %>%
  drop_na()

cat("Sample size with complete data:", nrow(rv_analysis), "\n")
cat("Mean RV burden:", round(mean(rv_analysis$RV_Burden), 2), "\n")
cat("Max RV burden:", max(rv_analysis$RV_Burden), "\n\n")

rv_model <- glm(AS_Binary ~ RV_Burden + Age + Sex + CAD_Binary,
                family = binomial(link = "logit"),
                data = rv_analysis)

rv_coef <- summary(rv_model)$coef["RV_Burden", ]
rv_or <- exp(rv_coef["Estimate"])
rv_se <- rv_coef["Std. Error"]
rv_pval <- rv_coef["Pr(>|z|)"]
rv_ci_lower <- exp(rv_coef["Estimate"] - 1.96 * rv_coef["Std. Error"])
rv_ci_upper <- exp(rv_coef["Estimate"] + 1.96 * rv_coef["Std. Error"])

cat("Rare Variant Burden Test Results:\n")
cat("Test: Logistic regression of RV burden on AS status\n")
cat("Adjusted for: Age, Sex, CAD status\n")
cat("Significance level (Bonferroni-corrected):", round(0.05/19, 6), "\n\n")

cat("Effect of RV Burden on AS Risk:\n")
cat("  OR (per additional rare allele):", round(rv_or, 4), "\n")
cat("  95% CI:", round(rv_ci_lower, 4), "-", round(rv_ci_upper, 4), "\n")
cat("  P-value:", format(rv_pval, scientific = TRUE, digits = 3), "\n")
cat("  Significant (Bonf):", ifelse(rv_pval < 0.05/19, "YES", "NO"), "\n\n")

cat("Full Model Summary:\n")
print(summary(rv_model))
```